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(1) Research Program/Project/Study Title: Acceptability of Semi-Dried Milkfish Tapa	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	ROXAS CITY
		Province:	CAPIZ
		Region:	VI
(5) Classification Research: ☒ Basic ☐ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity		(8) Discipline/Field of Specialization Food Engineering	
(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) (9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.) Milkfish, Tapa			

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(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) Tapa is dried or cured beef, pork, mutton, venison or horse meat, although other meat or even fish may be used. Filipinos prepare tapa by using thin slices of meat and curing these with salt and spices as a preservation method. When it comes to Tapa, one common concern revolves around the type of meat used, such as beef or pork, which can contribute to high cholesterol levels in our bodies. This aspect poses a challenge as high cholesterol is associated with various health risks. The issue lies in finding healthier alternatives or modifying the recipe to reduce the potential negative impact on cholesterol levels. Addressing these problems could involve exploring leaner cuts of meat, opting for poultry or seafood option or even considering plant-based alternatives to create a healthier version of meat tapa that retain sits delicious flavor while promoting a more balanced diet and overall well-being. Milkfish is a good source of protein. Consuming milkfish can meet the protein needs of the body. The high nutrient content in milkfish can prevent coronary heart disease, also increase endurance. Regular consumption of milkfish can prevent micronutrient deficiencies and can help the overall development of the body. So, include milkfish in your regular diet to obtain all the required nutrients for proper functioning of the body. Aside from its nutritional benefits, Milkfish is cheaper compared to other meats. According to a news article in Rappler, as of November 14, 2022, price watch, the price of Milkfish is Php 180.00/kilo while a kilo of beef is Php 360.00. It is obvious that its cost has given a significant value to a family's budget. This prompted the researchers to conduct an experimental study to create a fish-based meat product in the form of tapa as opposed to traditional pork meat-based products in order to prove that Milkfish or Bangus is the best alternative in making tapa as it provides a lot of benefits compared to other meats. The nutrients from Milkfish are beneficial to well-being as it is loaded with omega 3 fatty acids and protein which are quite essential for proper functioning of the body.
(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.) Materials The ingredients used in the study were the following: 2500 grams Milkfish (deboned), 60 grams salt, 60 grams sugar, 22.76 grams vetsin (MSG), 170.4 grams soy sauce, 170.4 grams vinegar, and 33.72 grams garlic chopped. Methods In all treatments, Milkfish were used along with other ingredients like salt, sugar, soy sauce, vinegar, garlic chopped and vetsin (MSG). Procedure: Preparation of Milkfish (Deboned). The milkfish were washed thoroughly under running water, then scaled and eviscerated. In eliminating the fishy odor, it was soaked in water with salt for 5 minutes. Drained and cut into four parts then removed the belly part. Preparation of Milkfish Tapa: 1. In a large mixing bowl, add-in the milkfish(deboned) 2. Add-in the black pepper, minced garlic, sugar, salt and vetsin (MSG). Massage the ingredients into the Milkfish and make sure it is evenly distributed. 3. Add-in the vinegar and soy sauce. 4. Mix the ingredients well and make sure the Milkfish is completely soaked in the marinade. 5. Cover the bowl and refrigerate for 24 hrs. 6. Sundried for 4 hrs., packed and put in freezer.
(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.) Experimental Design The study used experimental method of research. It is the method and procedure involving the control, manipulation of condition for the purpose of studying the relative effects of various treatments applied to the members of different samples. (Calderon, 2002). Experimental Treatment Four (4) treatments with two (2) replications were used in this study. Treatment A contained 25g Semi-Dried Milkfish, Treatment B contained 50g Semi-Dried Milkfish, Treatment C contained 75g Semi-Dried Milkfish, and Treatment D contained 100g Semi-Dried Milkfish. Details is shown in Table 3.1.

Table 1. Distribution of Ingredients per Treatment

Ingredients	Treatment A ew students	Treatment B	Treatment C	Treatment D Ge
Milkfish (Semi-Dried)	25g	50g	75g	100g
Salt	15 grams	15 grams	15 grams	15 grams
Sugar	15 grams	15 grams	15 grams	15 grams
Vetsin (MSG)	5.69 grams	5.69 grams	5.69 grams	5.69 grams
Soy Sauce	42.6 grams	42.6 grams	42.6 grams	42.6 grams
	42.6 grams	42.6 grams	42.6 grams	42.6 grams
Vinegar				
Garlic Chopped	8.43 grams	8.43 grams	8.43 grams	8.43 grams

Data Collection

The data was collected by providing the sample product to the respondents, who were provided with evaluation sheets containing scores cards with a 9-point hedonic scale. These sheets were used to assess the sensory attributes: color, odor, texture and taste. A total of 10 trained panelist were selected for the sensory evaluation of the product. A 9-point Hedonic scale were given to One Hundred (100) consumers to test the General Acceptability of the most preferred formulation (MPF) of Semi-Dried Milkfish Tapa. The researchers promptly collected the data and tabulated the results after the respondents completed the evaluation sheet.

Statistical Tools and Analysis

The statistical tools used in this study were mean, standard deviation, and one-way analysis of variants. Mean was used to determine the average scores in the results. Standard Deviation were used to determine the significant difference among the four different levels of maturity. The data was analyze using ANOVA (Analysis of Variance).

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Table 2 shows that the result of the rank test of Four (4) different treatment of Semi-dried Milkfish. Treatment A (25g Milkfish) on the first trial got the mean of 2.9 and a total of 2.2 on the second trial. Overall, the formulation got a mean rank of 2.55. Treatment B (50g Milkfish) on the first trial got a total of 1.8 and a total of 2.8 on the second trial. Overall, the formulation got a mean rank of 2.3. Treatment C (75g Milkfish) on the first trial got a total of 2.5 and a total of 1.9 on the second trial. Overall, the formulation got a mean rank of 2.2. Treatment D (100g Milkfish) on the first trial got a total of 2.5 and a total of 2.7 on the second trial. Overall, the formulation got a mean rank of 2.6

Based on the mean rank Treatment C had the highest rank 2.2 (75g Milkfish) which makes it the most preferred formulation, followed by Treatment B, then Treatment A and the least preferred formulation was Treatment D. The result shows that (75g Milkfish) of Milkfish meat is the most preferred formulation since it got the highest rank.

Table 2: Rank Test of the most preferred formulation (MPF) of Semi-Dried Milkfish Tapa

	Trial 1	Trial 2	Mean Rank
Treatment A	2.9	2.2	2.55
Treatment B	1.8	2.8	2.3
Treatment C	2.5	1.9	2.2
Treatment D	2.5	2.7	2.6

Table 3 shows that at 5% level of significance the result of Friedman test shows that there was no significant difference of the rank of 4 treatments. Thus, making all the formulation acceptable as the Most Preferred Formulation (MPF). The researchers based the result from the mean ranks for the MPF, which is why Treatment C was declared as the Most Preferred Formulation MPF for this study.

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Table 3. Friedman’s Test Results of the 3 formulations

N	20
Chi-square	1.800
Df.	3
Asymp. Sig.	0.615

Table 4 shows the general acceptability of the Four (4) different treatments of Semi-dried Milkfish. Treatment A (25g Milkfish) on the first trial had a score of 6.6 and got 8.4 on the second trial. Overall, the treatment had a mean of 7.15 which is described as “moderately acceptable”. Treatment B (50g Milkfish) on the first trial had a score of 8.2 and got 6.4 on the second trial. Overall, the treatment had a mean of 7.3 which described as “moderately acceptable”. Treatment C (75g Milkfish) on the first trial had a score of 7.3 and got 7.8 on the second trial. Overall, the treatment had a mean of 7.77 which is described as “moderately acceptable”. Treatment D (100g Milkfish) on the first trial had a score of 7.7 and got 7.1 on the second trial. Overall, the treatment had a mean of 7.4 which described as “moderately acceptable”. Based on the results, Treatment C had the highest acceptability level of 7.55 followed by Treatment D, then Treatment B and lastly Treatment A with the verbal interpretation of moderately acceptable.

Table 4: General Acceptability of Different Treatment of Semi-Dried Milkfish Tapa

Treatment	Trial 1	Trial 2	Mean	Verbal Interpreting
A	6.6	7.7	7.15	Moderately Acceptable
B	8.2	6.4	7.3	Moderately Acceptable
C	7.3	7.8	7.55	Moderately Acceptable
D	7.7	7.1	7.4	Moderately Acceptable

Table 5 shows that the Sensory Acceptability of Milkfish Belly In terms of Color, Treatment C (75g Milkfish) on the first trial had a score of 10.79 and got 11.32 on the second trial. Overall, the color of the most preferred formulation of Semi-Dried Milkfish Tapa with a mean of 11.06 which described as “Umber Brown”.

In terms of odor, Treatment C (75% Milkfish) on the first trial had a score of 8.71 and got 9.43 on the second trial. Overall, the odor of the most preferred formulation of Semi-Dried Milkfish Tapa with a mean of 9.07 which described as “Mild Fishy Odor”.

In terms of Texture, Treatment C (75g Milkfish) on the first trial had a score of 10.91 and got 10.12 on the second trial. Overall, the texture of the most preferred formulation of Semi-Dried Milkfish Tapa had a mean of 10. 52 which described as “chewy”.

In Terms of Taste, Treatment C (75g Milkfish) on the first trial had a score of 9.97 and got 9.19 on the second trial. Overall, the taste of the most preferred formulation of Semi-Dried Milkfish Tapa had a mean of 9.58 which is described as “Slightly Salty”.

Therefore, the sensory characteristics of Semi-Dried milkfish Tapa in terms of color was “amber brown”, in terms of taste was “slightly salty”, in terms of odor was “mild fishy odor”, and in terms of texture was “moderately soft”.

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Table 5 Sensory Characteristics of most acceptable treatment of Semi-Dried Milkfish Tapa in terms of color, odor, texture and taste

Sensory Characteri stics	Trial 1	Trial 2	Mean	Verbal Interpretin g
Color	10.79	11.32	11.06	Umber brown
Odor	8.71	9.43	9.07	Mild fishy Odor
Texture	10.91	10.12	10.52	Moderately soft
Taste	9.97	9.19	9.58	Slightly Salty

Results for the consumer test for general acceptability of the Most Preferred Formulation (MPF) of Semi-Dried Milkfish Tapa shown in Table 6. The table showed that out of One Hundred (100) respondents divided into four (4) categories/ untrained panelist which consist of Twenty-Five (25) Middle Age Adult with age ranging from 40-59, Twenty-Five (25) Adult with age ranging from 20-39, Twenty-Five (25) Teens with age ranging from 13-19, Twenty-Five (25) children with age ranging from 5-12, researchers found out in level of consumer acceptability of the MPF.

In middle adult category, 13 or 13% of them had rated the product as like extremely, 10 or 10% of them rated the product as like very much and 2 or 2% rated like moderately with the mean of 8.44.

In adult category, 17 or 17% of them rated the product like extremely, 6 or 6% of them rated the product as like very much, 2 or 2% of them rated the product as like moderately with the mean of 8.6.

In teens category, 14 or 14% of them rated the product as like extremely, 9 or 9% of them rated the product as like very much, 2 or 2% of them rated the product as like moderately with the mean of 8.48.

In children category, 13 or 13% of them rated the product as like extremely, 7 or 7% of the rated the product as like very much, 4 or 4% of them rated the product as like moderately, 1 or 1% of them rated the product as like slightly with the mean of 8.28.

Therefore, the target population of the study were adults and teens based on the data we gathered since this group of consumers rated the sample as “Like Extremely”.

Table 6: Consumers test for general acceptability of the most preferred formulation

	Child (5-12 yrs old)	Teen (13-19 yrs old)	Adult (20-39 yrs old)	Middle age adult (40-59 yrs old)
Mean	8.28	8.48	8.6	8.44

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate hedings if they are long. Justify paragraphs with main headings in caps.)

Conclusion and Recommendations

Tapa is dried or cured beef, pork, mutton, venison or horse meat, although other meat or even fish may be used. Milkfish as the main ingredient in Semi-dried Milkfish Tapa has a source of bioactive peptides that could have potential health benefits. The study found that milkfish proteins contain a variety of peptides that have antioxidant, antimicrobial, and anti-hypertensive properties, which could make them useful for developing functional food products (Journal Aqua-culture Research, 2018). Semi-dried milkfish retains many of the nutritional benefits of fresh milkfish,

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including being a good source of protein and essential amino acids. It also contains high levels of omega-3 fatty acids, which have been shown to have numerous health benefits (Saechan et al., 2021).

This study aimed to determine the acceptability of Semi-Dried Milkfish Tapa. This study was conducted at Capiz State University- Dayao Satellite College. Specifically, this study aimed to seek the following concerns: (1) determine the most acceptable treatment of Semi-Dried Milkfish Tapa (2) determine the sensory characteristics: color, odor, taste and texture of the most acceptable treatment of Semi-Dried Milkfish Tapa (3) determine the consumer acceptability of the most acceptable treatment of Semi-Dried Milkfish Tapa. One Hundred (100) evaluators rated the four formulations. Based on the data gathered, Treatment C (75g Milkfish) was the most preferred formulation. It had a general acceptability of “moderately acceptable”. In sensory attributes, in terms of Color it was rated as “Umber Brown”, in terms of Odor it was rated as “Mild fishy Odor”, in terms of Texture it was rated as “moderately soft”, and in terms of Taste it was rated “Slightly Salty”.

The target population of the study were adults and teens based on the data we gathered since this group of consumers rated the sample as “Like Extremely”.

Recommendations

In consideration with the findings observed in this study, the following recommendation were drawn:

1. Conduct Shelf-life of Milkfish Tapa by monitoring its microbial load, moisture content and water activity (Aw)
2. Determine the nutritional composition of Milkfish Tapa
3. Determine the right packaging materials.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Semi Dried Milkfish Tapa

(15) Technology Transfer Activity (Indicate the specific products, activity design, processes or services transferred as extension activities)

Production of Milkfish Tapa

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.)

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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